

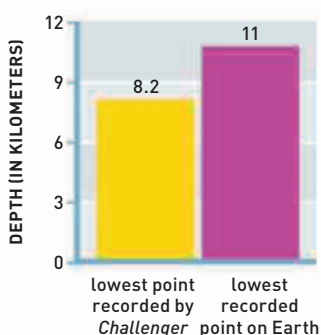
“ I THINK **LEPROSY** TO BE **INOCULABLE**; I, MOREOVER, THINK THAT **LEPROSY IN MOST CASES IS TRANSFERRED BY INOCULATION.** ”

Gerhard Hansen, Norwegian physician, from a letter to British social reformer William Tebb, c.1889

This micrographic image shows rod-shaped cells of the leprosy bacterium *Mycobacterium leprae*—the first bacterium to be identified as a cause of disease in humans.

IN 1872, AUSTRIAN PHYSICIST LUDWIG EDUARD BOLTZMANN (1844–1906) developed an equation that described the **behavior of a fluid** (gas or liquid) by applying probability distributions (see pp.1652–54) to the interactions of large numbers of atoms or molecules. This equation gave a **mathematical foundation** to the **second law of thermodynamics**, which states that systems tend toward a state of equilibrium.

In December 1872, a Royal Society expedition set out on board **HMS Challenger** from Portsmouth, UK. Over the next four years, it discovered many large features of the planet, including the mid-Atlantic ridge in the Atlantic Ocean and the Marianas Trench in the western



Challenger Deep

HMS Challenger estimated the depth of Earth's lowest point—Challenger Deep—at 5.1 miles (8.2 km). Recent estimates suggest 6.8 miles (11 km).

Pacific, where it took samples from the Challenger Deep—a point near the deepest recorded point on Earth. The expedition also catalogued approximately 4,700 previously unknown species of animals and plants.

The following year, Dutch physicist **Johannes Diderik van der Waals** (1837–1923) derived an “equation of state,” which described how the liquid and gas states of a substance merge into each other. He assumed that molecules exist, are of finite size, and attract each other by a weak force—now called a **van der Waals force**. Together, these ideas helped provide a better understanding of atoms.

In 1873, Scottish physicist **James Clerk Maxwell** published *A Treatise on Electricity and Magnetism*, in which he described his theory of **electromagnetism** (see pp.234–35). The theory predicted the existence of radio waves (see 1886), and was a major influence on 20th-century science.

William Crookes invented his **radiometer** while investigating the nature of light as a form of electromagnetic radiation. This device is a partially evacuated, airtight glass bulb that contains a set of vanes mounted on a spindle, like a horizontal windmill. One side of each vane is painted white, while the other is black.

When the vanes are exposed to light, the “light mill” rotates, with the white side leading the way. This occurs because the dark side of the paddle absorbs more radiant energy and gets hotter than the white side; some of the energy is transferred to the molecules hitting the surface, giving the paddle a kick. The same thing happens on the white side, but to a lesser degree.

The understanding of disease took a step forward in 1873, when Norwegian physician **Gerhard Hansen** (1841–1912) discovered the **leprosy** bacterium—*Mycobacterium leprae*. It was previously thought that leprosy was either inherited or spread by “bad air” known as miasmas.

Meanwhile, the mechanism of heredity was starting to be understood, partly through the work of German zoologist **Anton Schneider** (1831–90). He gave the first accurate

First practical refrigerator

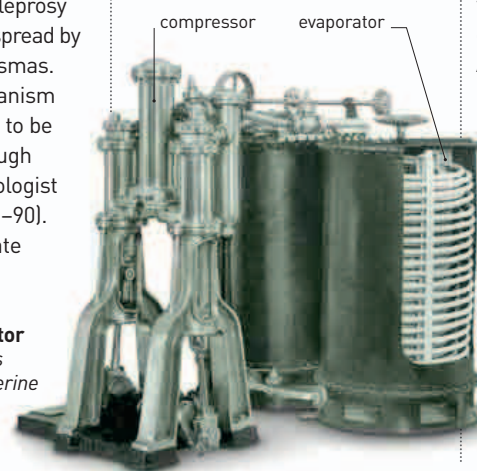
Von Linde improved on his 1873 design by using glycerine to seal the compressor and using ammonia as a refrigerant.



HMS Challenger

Launched on February 13, 1858, HMS Challenger was primarily a sailing ship, but was also fitted with an auxiliary steam engine.

description of **mitosis**—the process whereby a dividing cell provides identical copies of its chromosomes (then known as nuclear filaments) to each of the daughter cells (see pp.194–95).



The **first modern refrigeration system** was designed by German engineer **Carl von Linde** (1842–1934) and built by the mechanical engineering company Maschinenfabrik Augsburg for a brewery in Munich, Germany. Three years later, Linde designed a more reliable system—the first practical compressed-ammonia refrigerator. The huge commercial success of this invention enabled von Linde to focus on research, becoming the first person to **liquify air**, and **separate oxygen and nitrogen from it**.

1872 Eugen Baumann rediscovers **polyvinyl chloride (PVC)**

1872 Hungarian physician Moritz Kaposi describes the lethal cancer now known as **Kaposi sarcoma**

December 21, 1872 HMS Challenger sets out on a four-year-long scientific expedition from Portsmouth, UK

1873 James Clerk Maxwell publishes *A Treatise on Electricity and Magnetism*

June 14, 1873 Johannes Diderik van der Waals presents his thesis on the equation of state of liquids and gases

1873 Gerhard Hansen discovers the leprosy bacterium

1873 Anton Schneider observes the behavior of **chromosomes** during cell division

1873 Carl von Linde invents the first modern refrigeration system